

Adjusted CV — Andrei Kharlanchev

Andrei Kharlanchev connect.w.me.647@gmail.com | +54XXXXXXXXX | github.com/AndyHarman-dev

Senior UE5 gameplay and systems programmer with 4+ years of commercial Unreal Engine experience. Specializes in C++ architecture, data-driven gameplay systems, UI engineering, and performance-critical optimization across shipped titles.

Skills

Unreal Engine (UE5): UMG, Slate, Blueprints, Behavior Trees, Chaos Physics, Niagara
Languages: C++20, STL, Python, HLSL, RESTful API, C#, Java **Tools:** Git, Unreal Insights
Methodologies: Clean Code, SOLID, Agile/Scrum, Data-Driven Design, Design Patterns

Experience

Senior UE Developer, Lunas inc. – Montreal, Quebec, Canada *December 2025 – February 2026*

- Optimized Windows build by refactoring UI with a view pattern and introducing an event handling architecture driven by event-driven logic, cutting FPS drop by 97% (from 60 to 2 FPS).
 - Engineered a native Unreal host enabling .NET integration to leverage a free CAD library, and designed a full CAD-to-UE data conversion pipeline bridging two runtimes.
 - Engineered UStructSynchronizer to sync Unreal USTRUCTs with C# counterparts, achieving 63% reduction in data sync latency and ensuring data consistency across the .NET boundary.
 - Streamlined UI system by implementing a widget reuse pattern, reducing widget creation cost by 74% and eliminating per-interaction allocation spikes.
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Lead Senior UE Developer, byTalent – Sochi, Krasnodar Region *April 2024 – December 2025*

- Architected data-driven structures and flows in UE5 to represent 3D apartment objects, and built a custom packaging system that resolved asset dependencies via the Content Browser, categorized them automatically, and deployed the resulting .pak files to Amazon S3.

- Integrated REST API-based user authentication and a custom analytics tracking system recording dwell time per apartment and per-object interaction events, aggregating the data into structured reports dispatched to a backend endpoint.
- Authored custom Unreal Engine commandlets in C# to consolidate a multi-step manual terminal workflow into a single automated entry point with integrated error reporting, reducing deployment wait time by ~70%.
- Crafted widgets using the Slate Framework to bypass UMG's invalidation overhead, and applied draw caching in custom drag logic to avoid redundant element redraws.
- Produced utility modules reducing boilerplate in threaded operations, including Blueprint-friendly async wrappers, Rust-style result handling, and enhanced automation testing.
- Designed UI system from scratch using Presenter pattern for separation of concerns. Created customizable, reusable components matching Figma designs with system-wide configuration support.
- Assembled a cross-platform marketplace widget with platform-specific layouts and reusable components (filters, search, list selectors).
- Led a team of four developers, assigning tasks, managing disputes, and maintaining project coordination through an internal task tracker. Conducted biweekly progress meetings and architectural discussions. Wrote and maintained documentation for company guidelines and production pipelines.
- Resolved critical performance degradation during level loading and world streaming by refactoring to asynchronous asset loading and introducing object pooling, cutting FPS drops from ~45 frames (60 to 15–16 fps) down to 2–5 frames (60 to 55–58 fps).

Lead Middle UE Developer, Engagement – Krasnodar, Krasnodar Region *December 2022 – April 2024*

- Integrated a REST API to fetch product data with multi-variant structures, translating JSON responses into a product configurator UI that reactively filtered available options — colour, size, material — based on the current selection state.
- Engineered a cascading variant filtration system where each selection progressively narrowed subsequent options, ensuring only valid variant combinations were surfaced to the user.
- Established two caching mechanisms: one memory-based using a singleton for generic runtime data, and one file-based for byte-convertible cloud assets such as dynamic meshes, cutting asset load times by over 50% where no caching had existed previously.
- Refactored furniture moving system by isolating movement logic into testable components, decoupling input handling, and applying dependency injection, reducing the surface area for regressions when adding new interaction types.

- Built grouped object system mirroring Unreal Engine's actor grouping model, allowing users to select, move, and lock multiple furniture objects as a single unit with serialized group state.
 - Led a team of four developers, assigning tasks based on their skills and providing support when needed. Adopted Agile and Scrum practices to create a customized task pipeline, improving communication between developers and managers.
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Gameplay Programmer, Freelance *May 2022 – November 2022*

- Constructed destructible gameplay objects using Unreal Engine's Chaos physics system and Niagara effects, enabling multi-layered damage systems where objects like cars and buildings could be broken into progressively smaller pieces.
 - Built village gameplay systems on an RTS template, implementing dynamic weather, AI-driven NPC interactions and behaviors, and unit progression systems.
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Game Developer — Underestimated (indie project) — April 2022

- Programmed AI behavior systems using Unreal Engine Behavior Trees in a team of 4–5 developers; first experience with multi-person team synchronization and task coordination.

Game Developer — Drift (indie project) — March 2022

- Designed and shipped an inventory system inspired by Death Stranding, including full UI, drag-and-drop interaction, and inventory component logic in Blueprints.

Personal Projects — January 2022 – May 2022

- **Parkour Game** — solo UE5 prototype; established core movement mechanics and gameplay systems within first 2 months of learning the engine.
 - **Supernatural-inspired RPG** — inventory system, animation pipeline (Cascadeur), stealth mechanics, AI interactions, and spell/weapon systems in Blueprints.
 - **FPS Combat System** (private) — full C++ rewrite of a combat system with FPS/TPS perspectives; dedicated exercise to build C++ and engine architecture fundamentals.
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Projects

3D Game Engine — github.com/AndyHarman-dev/3D_Game_Engine

Portfolios

- Custom Details Config (UE Tools) — <https://github.com/AndyHarman-dev/UETooling/>

- TCP Poll Server — github.com/AndyHarman-dev/TCP-Polling-Server
- DI Container — github.com/AndyHarman-dev/DI-Container
- ISO 8583 Parser — github.com/AndyHarman-dev/ISO8583Parser
- TCP Auth Server — github.com/AndyHarman-dev/TCPAuthServer